

Manufacturer Response - Full Template Outputs

Current July_04 manufacturer response text.

Please use the attached Tapeout July_04 full template-output package.

It follows both supplied templates:

- VLSI-domain template: front-end, back-end, analog, mixed-signal, verification, physical verification, FPGA note, embedded systems, CAD/EDA, and fabrication-readiness outputs.
- PCB design-process template: schematic, BOM, footprints, netlist, board outline/mechanical, stackup/constraints, component placement, routing/planes, silkscreen/DRC/fab notes, Gerbers, DFA, and DFM outputs.

The schematic is included in `04\_front\_end\_schematic`. The dual NCE and TFLN view is in:

`04\_front\_end\_schematic/pdf\_svg/TFLN\_AI\_NODE\_X2\_CLEAR\_DUAL\_NCE\_TFLN\_SCHEMATIC\_OVERVIEW.pdf`

That overview identifies:

- U1 = NCE-A die
- U4 = NCE-B die
- U3 = TFLN_PIC_4xMZM photonic IC
- U50-U53 = RF driver chain
- U30-U35 and U40-U45 = HBM groups for the two NCE dies

Current closure state:

- KiCad DRC: 0 errors, 0 unconnected items, 82 library/footprint mismatch warnings
- Unconnected items: 0
- Schematic parity: 214 warnings requiring the complete schematic/netlist source
- BOM: current July_04 BOM review list; AVL, quantity, DNP/populate status, substitutions, and parity require owner signoff

Please use `00\_read\_me/ALL\_TEMPLATE\_OUTPUTS\_INDEX.pdf` as the top-level map and

`17\_release\_gate/RELEASE\_GATE\_CHECKLIST.pdf` as the release gate.